

Real biomedical data, reproducible analysis

NHANES 2017-2018 | Excel, Power Query and SQLite | 7 October 2026

Question

How do data availability and the distributions of six liver and kidney biomarkers vary by age and recorded sex in the adult NHANES sample?

Cohort and transformations

The biochemistry file contains 6,401 records. A checked one-to-one participant-ID join matches every laboratory record to demographics. Restricting to age 20 and older retains 5,265 adults and excludes 1,136 records. Reshaping six biomarkers produces 31,590 measurement slots, including missing values.

Checkpoint	Count
Laboratory records	6,401
Adults aged 20+	5,265
Observed measurements	29,479
Missing measurement slots	2,111
Imputed values / removed outliers	0 / 0

Cleaning decisions

Preserve the original XPT files and published measurement units. Normalize decoder artifacts only in source code/weight fields. Decode source-recorded sex and retain the 80+ age category. Preserve ALT/GGT detection-limit flags. Use a pooled adult 1.5-IQR rule to mark values for statistical review. No flagged measurements are removed or classified as clinical abnormalities.

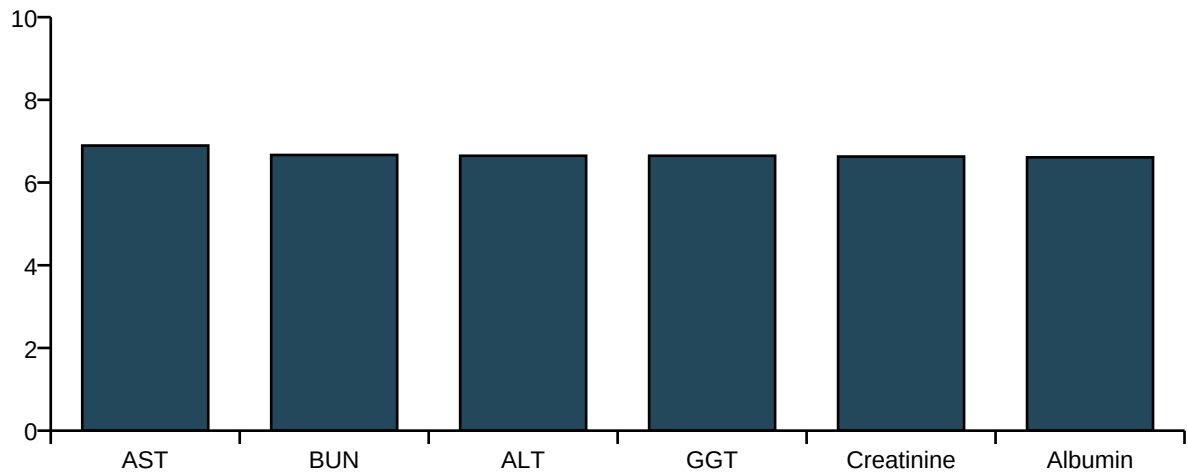
Scope

All results describe the included sample without survey weighting. They are not US population estimates, causal findings or diagnoses. Survey weights, strata and primary sampling-unit fields remain available for later design-aware work.

Availability and distributions

Missing measurements by biomarker

Unweighted adult sample. Missing values (%)



Biomarker	Observed	Missing	Missing %
AST	4,902	363	6.89%
BUN	4,914	351	6.67%
ALT	4,915	350	6.65%
GGT	4,915	350	6.65%
Creatinine	4,916	349	6.63%
Albumin	4,917	348	6.61%

Interpretation

Adult-sample missingness ranges from 6.61% for Albumin to 6.89% for AST. The six biomarker-specific denominators are all 5,265, because the long-form transformation retains missing slots. Differences in availability should be assessed before complete-case analysis or imputation.

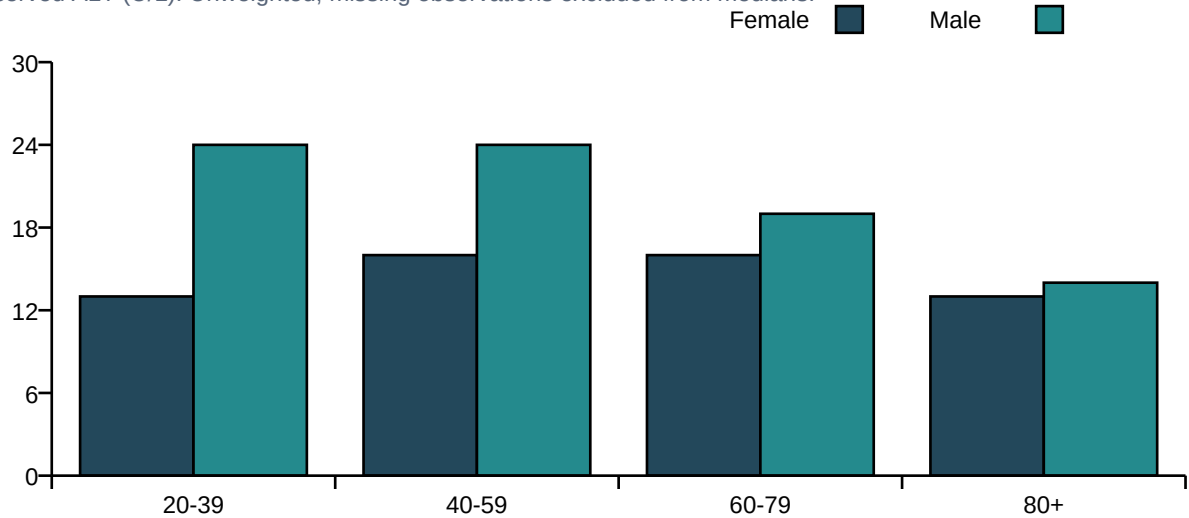
Counts matter

The grouped CSV includes the participant, observed and missing counts for every biomarker, age band and recorded-sex group. Medians and means exclude missing values. Do not average measurements across biomarkers with different units.

Audit and reproducibility

ALT medians by age and recorded sex

Observed ALT (U/L). Unweighted; missing observations excluded from medians.



Cross-tool checks

SQL JOIN/CTE/UNION ALL views independently reconstruct the cohort and long format. SQL window functions calculate group medians. The pipeline reconciles all 31,590 long-form records and 48 group summaries against these SQL results. SQLite integrity and foreign-key checks pass. CSV exports are read back and compared to their source frames.

Excel verification status

The native review workbook opened, recalculated, saved and reopened in desktop Excel. Six formula summaries and the age/sex selector results matched the verified data for ALT and Creatinine. Both charts persisted. Power Query opened, but editor text input through app control failed; script activation also returned access denied. Query refresh, PivotTables and slicers remain pending. Their M queries and completion script are supplied, with verification status documented.

Source and authorship

Source: CDC, National Center for Health Statistics. National Health and Nutrition Examination Survey, 2017-2018, Standard Biochemistry Profile and Demographics. Public-use data retrieved 7 October 2026. Source file hashes are pinned in the pipeline. This is an AI-assisted portfolio project; contributions and verification status are documented in the repository.

[CDC biochemistry codebook](#)

[CDC demographics codebook](#)